

Temporal Equivalence Principle: A Proper-Time Domain Boundary in the Outer Solar System

Matthew Lukin Smawfield

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Abstract

This paper investigates whether outer-Solar-System orbital structure motivates a domain-boundary test of the Temporal Equivalence Principle (TEP). The approach compares resident trans-Neptunian objects with transiting long-period comets, using public catalogues, selection controls and numerical orbit propagation. A secure detached sample of 44 objects shows a perihelion-longitude concentration with resultant $R_{\omega} = 0.332$ ($p = 0.00735$ under an angularly randomized, orbit-conditioned null). In 54 CODE comets absent from the Warsaw sample, original-to-future periapsis rotations are 0.212° inside the resident-derived sector and 0.131° outside ($p = 0.00295$). These rotations are also reproduced by standard planetary dynamics; their directional regression residual is therefore a population statistic, not a direct failure of the dynamical model. The proposed equivalent-time conversion is not a measurement of proper time.

The clearest additional spatial result is an aphelion-direction dipole toward the same sector in both SBDB eras: $d_{\parallel} = 0.110$ before 2018 and 0.0949 afterward. The latter survives both Galactic-latitude and ecliptic-latitude nulls ($p = 0.00215$ and 0.00225). In contrast, the temporally held-out comet cohort fails the predicted rotation excess. Subsequent bipolar geometry remains exploratory, and its modern residual is concentrated in mixed astrometric-catalogue observations. The combined evidence motivates a shared spatial hypothesis and identifies concrete tests for TEP, but does not yet distinguish a temporal boundary from selection and reconstruction effects. That distinction requires a specified matter-metric response, a survey-selection model, and successful prediction of observations withheld from model development.

Key words: Kuiper belt: general – comets: general – gravitation – celestial mechanics – astrometry

1. Introduction

1.1 The question posed by distant orbital structure

The apparent concentration of some extreme trans-Neptunian perihelia has motivated distant-planet models (Trujillo & Sheppard 2014; Batygin & Brown 2016). Its interpretation remains sensitive to discovery selection. Characterized-survey analyses have found that their extreme-object samples are consistent with an underlying unclustered distribution once selection is included (Shankman et al. 2017; Napier et al. 2021). The scientific question is consequently whether a common physical structure is required after the discovery and measurement processes have been modelled. This paper investigates a temporal-boundary hypothesis using both resident objects and comets passing through the outer system.

A second population offers a useful test because it has different trajectories, discovery histories and fitted observables. Agreement can expose a common organizing direction. It does not by itself identify the organizing mechanism: shared sky visibility, inherited astrometry and analytic choices can also correlate the evidence. Three levels of inference are therefore distinguished throughout: a feature of the observed catalogue; rejection of a specified statistical null; and preference for a physical TEP model over conventional alternatives.

1.2 TEP and a domain-boundary hypothesis

The canonical TEP framework couples matter to a metric

$$\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu} + B(\phi)\nabla_\mu\phi\nabla_\nu\phi,$$

where ϕ is a scalar field and A and B specify conformal and disformal responses. A localized outer-system transition in this field is the hypothesis under investigation. The boundary should be understood as a continuous field configuration; its position, width, amplitude, motion and environmental response must ultimately follow from a specified model. The statistical cap used here does not supply that solution.

In the conformal limit, along a specified timelike worldline, $d\tilde{\tau} = A(\phi) d\tau_g$. This relation explains why a spatially varying field could affect clock comparisons, but it does not equate an orbital-element angle with elapsed proper time. Matter trajectories, photon propagation, local reference clocks and the orbit-fitting procedure must be treated together. Moreover, ordinary general relativity already predicts path-dependent proper time; observing an orbital anisotropy cannot alone establish a new dynamical time field.

The attractive feature of the TEP hypothesis is its potential economy: one specified field could predict responses across orbital and clock channels. That economy becomes evidence only when the common parameters impose quantitative restrictions. Allowing a different sign, axis, epoch or screening factor after seeing each result removes the predictive advantage. The present analysis is a search for observational constraints on such a model, rather than a completed likelihood comparison between a derived TEP solution and standard gravity.

1.3 Resident and transit probes

The resident sample uses $a > 150$ AU and $q > 30$ AU; the more extreme subset uses $a > 250$ AU. These are operational catalogue cuts. Objects close to the perihelion threshold can still interact with Neptune, so residence in a proposed field region is not established simply by classification. Long-period comets provide a different probe through their angular elements, published original and future orbits, and independently fitted astrometric arcs.

The detached sample yields a direction near $(49.9^\circ, -17.0^\circ)$ in ecliptic J2000 coordinates; the more extreme subset yields approximately $(34^\circ, -13^\circ)$. The latter defines the principal comet cap. These are distinct, correlated estimates from nested resident samples. The analysis records both rather than treating every test as if it used one identical axis. “Resident-derived” denotes their construction; it is not a claim of an externally timestamped preregistration.

Sections 2–4 describe the data and compare the resident, transit and spatial channels. Section 5 addresses combined evidence and spacecraft controls. Section 6 states the physical and statistical tests required to turn the spatial hypothesis into a discriminating TEP prediction.

2. Data, observables and analysis design

2.1 Catalogues and independence

The empirical measurements are public observations and published orbit solutions. Simulated null catalogues, orbit integrations, bootstrap samples and power calculations are computational controls; they are not additional observations. The JPL Small-Body Database (SBDB) supplies resident and cometary elements. Warsaw and CODE supply osculating heliocentric and original/future barycentric comet orbits. The one-apparition catalogue supplies another comet sample with published element uncertainties and comparison solutions. MPCORB, CometEls, DES and the characterized OSSOS ensemble support catalogue and discovery-selection checks.

Table 1. Principal samples and their role.

Sample	Membership	Role and limitation
SBDB secure detached	44; $a > 150$ AU, $q > 30$ AU, condition code 0–3	Resident direction and clustering; heterogeneous discovery selection
SBDB extreme subset	26; $a > 250$ AU, otherwise the same cuts	Comet-sector definition; nested in the resident sample
CODE-only deep plungers	54 class-1 solutions; $q < 3.1$ AU, absent from Warsaw	Object-disjoint comparison with shared orbit-determination lineage
SBDB post-2017 spatial sample	287 near-parabolic comets	Aphelion directions; no successful boundary integration required
SBDB post-2017 primary rotation sample	121 successful integrations from 148 eligible deep plungers	Temporal holdout; 27 integration failures require selection accounting
DES detached	16, including 14 shared with the secure SBDB sample	Separate survey/reduction; largely overlapping objects
OSSOS detached class	31 under the survey's dynamical classification	Characterized discovery geometry; different orbital cuts

Different catalogue names do not imply independent data. Warsaw, CODE and the one-apparition catalogue share a methodological lineage. SBDB and MPCORB often describe the same bodies, while separate orbit fits can reuse the same astrometry. DES provides an independent survey and reduction but overlaps the secure resident membership. In the ISO analysis, arrival, periapsis and departure directions belong to one orbit and must be handled as a correlated group.

2.2 What is actually measured

The perihelion-direction unit vector $\hat{\mathbf{p}}$ follows from inclination i , node Ω and argument of perihelion ω . Longitude of perihelion $\varpi = \Omega + \omega$ is a circular angle; a cut on ϖ is distinct from a three-dimensional cap on $\hat{\mathbf{p}}$. A uniform circular distribution places one third of its members within a 60° half-width, whereas an isotropic sphere places one quarter within a 60° cap. Each null must match the statistic actually used. Angular elements are not clock readings.

For comets, $d_{\text{of}} = \angle(\hat{\mathbf{p}}_{\text{orig}}, \hat{\mathbf{p}}_{\text{fut}})$ denotes the original-to-future periapsis angle, with analogous original-to-osculating and osculating-to-future angles. The published original and future solutions are obtained by dynamical propagation to the outer boundary; they are not independent observations at 250 AU (Królikowska 2014). Their agreement with a second integrator validates propagation. It does not, without an observation-level comparison, validate a new force or a time anomaly.

The spatial statistic is $d_{\parallel} = N^{-1} \sum_i \hat{\mathbf{a}}_i \cdot \hat{\mathbf{n}}$, where $\hat{\mathbf{a}}_i = -\hat{\mathbf{p}}_i$ is the aphelion direction and $\hat{\mathbf{n}}$ the reference axis. This avoids differencing propagated boundary legs. It remains a quantity inferred from fitted orbital elements, subject to discovery selection and astrometric error: “spatial” does not mean independent of reconstruction.

2.3 Frames, numerical precision and nulls

Orbital directions are expressed in the ecliptic J2000 frame. The shared transformation to Galactic coordinates first applies the J2000 obliquity rotation to equatorial coordinates, then the equatorial-to-Galactic matrix. Directly applying an equatorial-to-Galactic matrix to ecliptic vectors is incorrect. The adopted transform is checked against Astropy and the SPICE frame convention. The resident and transit axes have different Galactic coordinates; the transit axis is approximately $(166^\circ, -54^\circ)$.

The conditioned resident null holds (a, e, i) fixed and randomizes the angular orientations. Footprint controls add assumptions about discovery longitude and its coupling to perihelion direction. Galactic-latitude and ecliptic-latitude comet nulls preserve selected latitude information while randomizing other orientation components. These are explicit and useful reference models, but neither substitutes for a complete population-plus-survey simulation.

Finite Monte Carlo tests use $p = (k + 1)/(B + 1)$, with k null statistics at least as extreme as the observed statistic among B simulations. Zero exceedances in 20,000 draws correspond to a reported rank of 1/20001, not a measured zero probability. Full precision is retained before combining tests. Historical result files include varied reporting conventions; only explicitly bound result leaves are used for the main quantitative claims.

3. The resident signature: spatial alignment

3.1 The measured concentration

The secure detached sample has $R_{\varpi} = 0.332$ and mean longitude 49.1° , with a 68 per cent bootstrap interval of 33.2° – 66.6° (step 010). Its conditioned angular-null probability is 0.00735. This is evidence against that specified reference distribution. The estimated direction and its uncertainty come from the same objects whose clustering is tested; subsequent axis-conditioned resident statistics are not fresh replications.

The extreme subset also concentrates in the same broad sector, but its smaller membership gives weaker evidence. The five-object high-perihelion subset is not significant under its corresponding conditioned null. Scans over detachment thresholds, angular cap sizes and orbital quality are useful sensitivity checks; nested cuts are highly correlated and their count cannot be interpreted as a count of independent detections (steps 023, 025 and 111).

3.2 What the footprint tests establish

The discovery-footprint analysis measures a proxy for each object's discovery longitude and explores its coupling to ϖ (steps 014–016). Setting perihelion longitude equal to that proxy gives a resultant of 0.295, below the observed 0.332. This deterministic realization is not a universal ceiling on discovery bias: finite random scatter can exceed a mean attenuation curve, and object-dependent coupling, magnitude, latitude, tracking and quality selection can all alter the distribution. A proper bound requires the full relevant sampling process, with its tail probability.

The footprint-conditional tests therefore narrow particular explanations rather than structurally excluding selection. The same qualification applies to the apparent decorrelation with discovery direction at large semimajor axis (step 017): a population transition in a proxy diagnostic, not a direct measurement of the radius of a field boundary. The characterized-survey literature supplies an essential opposing benchmark: simulated discovery selection can generate strong angular structure in intrinsically unclustered populations.

3.3 Survey-by-survey evidence

The resident ledger compares each survey cohort with its own adopted footprint baseline (step 113). SBDB has 26 of 44 objects in the longitude cap, compared with a baseline fraction near 0.42. DES has 13 of 16, compared with about 0.55. The broader OSSOS detached class has 13 of 31, compared with about 0.40, and is consistent with that baseline. These divergent outcomes are informative and are retained rather than averaged away.

Table 2. Resident evidence under the adopted footprint models (step 113).

Cohort	In-cap fraction	Interpretation
SBDB secure detached	26/44	Excess under the designation-based footprint proxy
DES detached	13/16	Excess under its proxy; 14 objects overlap SBDB
OSSOS detached class	13/31	Consistent with its discovery-astrometry baseline
Deduplicated union	39/76	Baseline comparison $p = 0.0803$

The deduplicated union is a useful check against double counting, although it combines different operational definitions of “detached”. Its baseline comparison is weaker than a nominal Fisher combination of overlapping survey entries. Later discovery-longitude windows and empirical coupling kernels reveal where the adopted mixture model distinguishes the samples (steps 115–116). Because the window, bandwidth, mixture and securing model were developed using the observed record, their likelihood ratios are conditional model comparisons, not posterior odds for TEP; the full procedure requires calibration on withheld survey data.

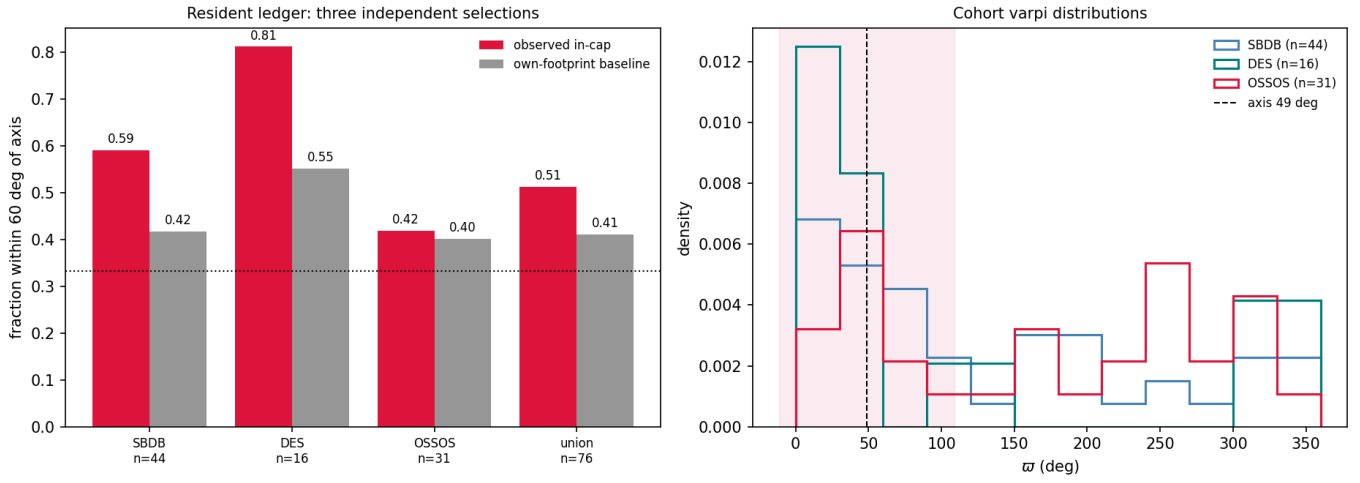


Figure 1. Survey ledger from step 113. Cohort-specific baselines and the deduplicated union are the relevant controls. Overlapping survey entries are not independent discoveries of a field.

3.4 Dynamical context

The pipeline examines element coupling, population selectivity, pole geometry and secular evolution (steps 011, 018–024, 066, 088 and 090). These tests constrain the particular dynamical realizations and initial conditions simulated. A lack of detected resonance structure in a small selected sample does not exclude every distant-planet population model. Likewise, a 20 Myr integration of the observed distribution is a stability experiment; it is not a comparison with a generative, survey-selected population evolved over Solar-System history.

The strongest resident result is therefore a measurable preferred sector under several explicit reference models. The remaining task is to determine whether the excess survives a survey simulator that includes the observed selection and quality cuts. This is a well-defined empirical target for any TEP field model.

4. Transit geometry and the spatial aphelion channel

4.1 A directional population contrast

Among the 54 CODE-only deep-plunging class-1 comets, 19 lie in the principal three-dimensional cap. Their median original-to-future periaapsis rotation is 0.212° , compared with 0.131° outside ($p = 0.00295$; step 034). The energy-kick comparison is not significant ($p = 0.688$). This is a directional contrast in the selected population's propagated orbital response; the nonsignificant energy comparison demonstrates neither zero exchange nor equivalence between energy distributions.

The Warsaw leg decomposition and the CODE comparison offer useful sample-level corroboration, while their common orbit-determination lineage limits methodological independence. Formal uncertainties, observational leverage, sky matching and published nongravitational solutions are examined in steps 058–060 and 079–082. These checks reduce specific confounds, but they do not propagate a full orbit covariance through the entire selection, integration and regression procedure, and pure-gravity fits do not establish the absence of physical cometary outgassing.

4.2 The full dynamical prediction and its reduced regression

The REBOUND/DE440s integrations reproduce the published original-to-future rotations closely (step 063). In the 54-object deep-plunger sample, the catalogue/simulation rank correlation is approximately 0.999 and the median absolute difference is about 6.4×10^{-4} degrees. The direct catalogue-minus-simulation cap test gives $p = 0.817$. By contrast, regressing the simulated rotation on a few summary covariates gives a cap residual with $p = 0.00188$. These are different tests with different meanings.

Write the standard-dynamics rotation as D_i^{dyn} , and a reduced regression on energy kick, sampled encounter distance, perihelion depth and inclination as $m(X_i)$. The statistic used in several pipeline steps is

$$r_i^{\text{proxy}} = \log D_i^{\text{dyn}} - m(X_i).$$

The observation-minus-prediction comparison would instead involve $D_i^{\text{obs}} - D_i^{\text{dyn}}$ with a specified observational covariance. An anisotropy in r_i^{proxy} shows that these summary covariates do not remove all directional variation from the selected standard-dynamics population, yet it cannot establish that the full planetary calculation leaves the same amount of unexplained rotation.

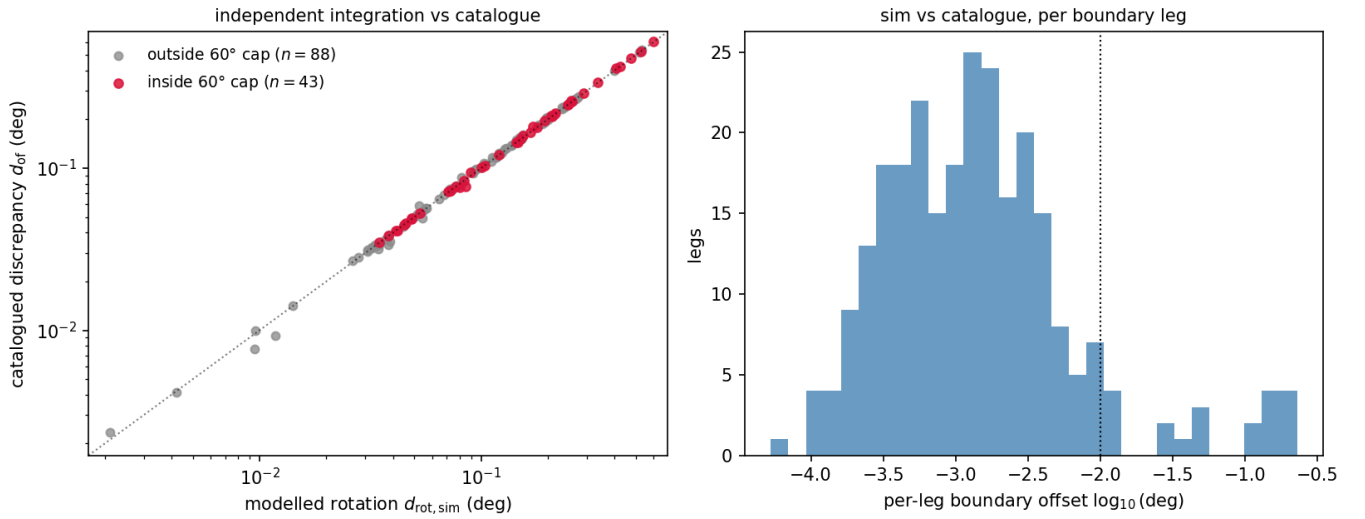


Figure 2. Bidirectional propagation comparison (step 063). Left: catalogue and standard-dynamics rotations in the full integrated sample of 131 objects. Right: the per-leg boundary-direction offsets between the integrated and catalogue solutions, displayed as $\log_{10}$ degrees. The 54 deep plungers discussed in the text are a subset of this sample.

The sampled minimum encounter distance is also an integration-output diagnostic, not necessarily the true continuous minimum of a close encounter. Boundary extraction, output cadence, nongravitational forces and covariance propagation require convergence checks before small per-object discrepancies can be interpreted. The catalogue agreement is valuable validation of the bulk machinery, but cannot replace those checks for a proposed new effect.

4.3 Equivalent orbital time is not a clock measurement

The pipeline converts an angular scale into years using

$$\Delta t_{\text{eq}} = \Delta \theta r_b^2 / h,$$

where h is specific angular momentum and r_b a chosen reference radius (step 065). Since h/r^2 is the true-anomaly sweep rate, this defines an equivalent orbital phase timescale. A pure Keplerian phase advance changes the position along an orbit while leaving the eccentricity-vector direction unchanged. A periapsis-direction rotation and a phase advance therefore cannot be identified without an additional response calculation.

The reported “slip” maps, their harmonic fits and cross-validation scores describe this transformed statistic (steps 068, 071, 073, 075–076, 086, 089 and 102–106). They can be used to compare empirical profiles, provided the reference radius is fixed and the inherited regression assumptions are retained; they do not yet measure accumulated proper time, a lapse amplitude, or a spatially localized clock-rate discontinuity, and the factor r_b^2 must not be mistaken for independent evidence of a radial field law.

4.4 What the raw-astrometry refits add

The independent differential-correction implementation fits raw MPC astrometry and propagates the resulting trajectories under the same standard force model (steps 127–128 and 134). It avoids Warsaw fit products, but shares observations, a catalogue seed and a gravitational model with other constructions. It is a useful test of dependence on one orbit-fitting implementation; it is not an independent measurement of time.

The stored pre-2018 record contains 587 fitted comets. Its full-sample rotation-regression comparison has $p = 0.184$; its inbound carrier has $p = 0.0340$; the independent-leg disagreement comparison has $p = 0.107$. Reporting only the smallest carrier probability would conceal that the observable choice matters. The smaller historical signature refit also illustrates this sensitivity: for 17 usable objects, the one-sided Mann–Whitney test gives 0.0409, while the median-gap permutation gives 0.0690. These test different distributional summaries.

The refits use a simplified astrometric response and clipping procedure, without a complete station/catalogue error model, orbit-covariance propagation, or a demonstrated detection efficiency for a specified TEP signal. Near-perfect correlation with an orbit seeded from the same observational record is not, on its own, certification that all reduction systematics are absent. The optimiser and cached results must be validated and versioned before very small fit-derived differences can be treated as precision evidence.

4.5 The temporal holdout fails the declared prediction

The primary post-2017 SBDB cohort was reserved relative to the historical CODE sample (step 117). Its 121 successful deep-plunger integrations give median rotation 0.114° inside the declared cap and 0.171° outside — the opposite of the predicted ordering. The prespecified greater-than comparison gives $p = 0.956$, and the full reduced-regression comparison gives 0.970. The 27 failed integrations among eligible primary objects enter the failure accounting; they are not silently treated as null outcomes.

This result does not validate the claim that the same fixed sector produces a stable positive rotation excess in new temporal cohorts. Later searches find a displaced modern direction and an era-dependent bipolar description (steps 118–123 and 125–133). Those analyses are useful for generating a revised hypothesis, but the new axis, epoch dependence and chosen morphology cannot retrospectively rescue the original held-out prediction. A revised model must be tested on new data.

4.6 The astrometric-catalogue control

Per-observation catalogue labels provide a direct test of a reduction-era confound (step 140). The modern displaced residual is nonsignificant in the uniform-Gaia stratum ($p = 0.283$) and much stronger in the mixed-catalogue stratum ($p = 6.45\text{e-}7$). The legacy inbound carrier remains significant within its corresponding stratum ($p = 0.00254$). Uniform-Gaia denotes an adopted composition threshold, not a guarantee that every observation is Gaia-reduced.

Matching average catalogue composition does not exclude an interaction between catalogue, sky direction, epoch and observing station. The stratum contrast therefore weakens a universal displaced-field interpretation and motivates a catalogue-stitching explanation, without proving that explanation or establishing that the legacy signal is free of all catalogue errors. This control must accompany any bipolar or moving-boundary claim.

4.7 The spatial aphelion result remains a distinct target

The spatial aphelion statistic yields a more persistent result. On the SBDB record, the pre-2018 sample of 880 comets has $d_{\parallel} = 0.110$, and the post-2017 sample of 287 has 0.0949 (step 124). Under the latitude-conditioned Galactic reference, their finite-simulation probabilities are 0.0000500 and 0.00215. The post-2017 ecliptic-latitude reference gives 0.00225. The older sample reaches the 20,000-draw resolution floor.

Because it does not depend on a successful original-to-future integration for each object, this persistence is the clearest reason to retain a common spatial hypothesis despite the failed rotation holdout. The CometEls census and its magnitude, era and perihelion controls provide additional checks (steps 108–110), with overlapping membership explicitly relevant to their evidential weight. These catalogues remain selected samples: preservation of one latitude distribution is not proof that all angular visibility and discovery effects have been removed.

The nominal two-era Fisher combination is 0.00000183 under independent, valid component nulls. The eras contain disjoint objects but share selection assumptions and an axis chosen from other observations; the combined number therefore summarizes the stated references rather than measuring a calibrated TEP probability, and the older era's Monte Carlo floor limits the precision of any combined tail estimate.

5. Synthesis and cross-channel constraints

5.1 An evidence ledger with distinct roles

The case for investigating TEP rests on a conjunction worth testing: the resident directions identify a sector; a selected historical comet sample has an angular-response contrast there; and the aphelion spatial statistic persists across comet eras. The case is limited by a null direct catalogue-minus-dynamics contrast, failure of the temporal rotation prediction, overlapping catalogues and incomplete survey-selection modelling. A persuasive account must explain both sides of that ledger.

Table 3. Evidential status of the main channels.

Channel	Result	What follows
Resident angular concentration	Conditioned-null excess	Tests selected angular reference models
Historical comet rotation	In-cap population excess	Standard dynamics also reproduces the rotations
Temporal rotation holdout	Predicted excess fails	Fixed-sector, fixed-sign prediction is unsupported
Spatial aphelion dipole	Positive in both eras	Persistent target for a survey-calibrated spatial test
Modern bipolar interpretation	Catalogue-stratum dependence	Exploratory; systematics remain a live explanation
Spacecraft clocks and RTGs	Calibration/degradation-limited diagnostics	Model-dependent constraints, not independent field detections

The historical two-population Fisher combination is 0.000255 using the exact resident and CODE source values (step 051). It is nominal under independence and the validity of both component nulls. Within-family false-discovery corrections on a selected historical test list do not account for every axis, cut, outcome, regression specification and explanatory family explored in the full pipeline. Those exploratory results should not be compressed into a single discovery significance.

5.2 What a random-axis landscape measures

Step 126 combines ten directional summaries into a score. At the resident analysis axis its current value is 191.4, exceeded by a fraction 0.00890 of 20,000 random directions. This describes how localized the observed score landscape is while the catalogues are held fixed and only the test direction changes. Because it does not generate new null datasets and repeat the entire axis-selection procedure, it is not a global look-elsewhere correction.

Several resident score terms first select objects inside an angular cap and then apply an unrestricted Rayleigh formula. Such selection itself concentrates angles, so those terms are descriptive weights rather than valid component probabilities. The catalogues also overlap, so evaluating them at shared directions preserves features of their observed correlation without making the ten terms independent or validating their nulls. A valid global test would simulate or resample under an explicitly justified null, preserve object identities and survey dependencies, repeat all fitting and searching, and compare the resulting maximum score with the observed one.

5.3 CMB-frame and mirror comparisons

The subsequent CMB-frame analysis examines opposite era dipoles, cone bands, a meridian plane and a reflected lobe (steps 129–135 and 139). The CMB direction is physically external to this comet dataset, but selecting which relation to test against it can still be retrospective. In particular, freezing a plane or mirror rule after inspecting these same directions does not create a prospective experiment.

The displaced direction lies about 11.5° from the equatorial mirror of the declared direction. Preserving its measured cone and randomizing its azimuth gives a tail of 0.0733. The corrected joint meridian test jointly rotates the three directions of each ISO and excludes the resident direction that informed the sector; its conditional reference tail is 0.0775. Neither reaches a 0.05 threshold. These calculations describe known geometry and do not include the full history of selecting the geometric hypothesis.

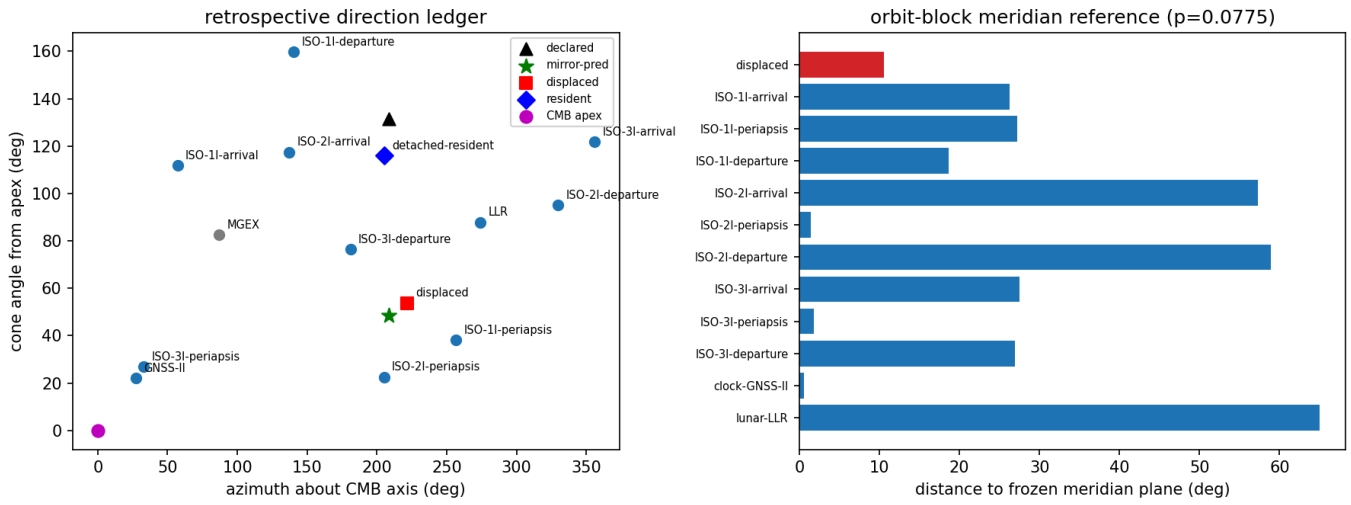


Figure 3. Retrospective geometry audit (step 139). The historical filename is retained. The joint reference preserves the orbital dependence of each ISO's arrival, periapsis and departure directions. Training-related and non-identifiable directions are excluded from the combined score.

The GNSS-II direction is close to the proposed meridian, whereas the freely recovered LLR direction is far from it. A product of their plane-distance probabilities is not itself a joint p-value; the corresponding nominal Fisher reference is 0.0523. The broader cross-channel Fisher calculation uses one GNSS-II statistic rather than counting that same direction twice, and reads the actual comet result instead of substituting a constant when a result key is missing. Its value, 8.97e-9, remains nominal: that arithmetic does not resolve shared GNSS observations, directional sensitivity, source-test calibration, or the retrospective choice of relations.

5.4 Spacecraft, plasma, nuclear and ISO channels

The spacecraft analyses use NAIF spacecraft-clock correlation kernels, Voyager plasma-wave data and RTG electrical-power records (steps 095–100, 103 and 137–141). NAIF SCLK coefficients map spacecraft clock counts to a reference time system. Changes in a piecewise clock-correlation solution cannot be identified directly with an anomalous oscillator frequency without modelling the correlation and operational procedures.

The signed crossing diagnostics contain negative fitted steps, but their comparison windows overlap and two crossings on one spacecraft are not independent experiments. Comparing a marked epoch with alternative epochs in the same time series gives a descriptive rank unless time-exchangeability and the selection of windows are justified. Independent-crossing binomial or Fisher references are therefore illustrative only. A cancellation property of a static propagation term in an idealized coherent two-way link does not automatically remove endpoint-clock, motion, calibration or time-dependent contributions from every measured channel.

Plasma-wave frequencies depend on electron density as well as any proposed measurement response; an environmental boundary is not independently a time-field boundary. RTG electrical power depends on thermoelectric conversion and aging as well as radioactive heat. The cross-generator control is valuable, but sibling units share bus, thermal and operational history. Converting electrical-power scatter into a constants or lapse bound requires explicit nuclear and engineering response assumptions; it cannot be called a direct Pu-238 clock measurement.

The three ISOs offer geometrical examples rather than a large statistical sample (step 101). Their nongravitational accelerations and explanatory classifications are heterogeneous, and a three-object classification match is sensitive to reclassifying one object. The ISO orbit directions are already represented as dependent groups in the geometric audit. No ISO geometry match is counted as an independent confirmation of TEP.

The solar-wind epoch analysis supplies a useful negative control (step 136): the pooled pressure association does not persist within the eras. Contemporaneous pressure at the adopted Earth-based proxy consequently does not support the proposed screening mechanism. This does not test every delayed, spatially transported heliospheric response, which would require its own specified model.

6. Physical interpretation and decisive tests

6.1 What conventional alternatives have and have not been excluded

Direct perturber insertions constrain the particular masses, orbits, phases and integration intervals tested (steps 062, 083, 087, 091 and 093). They show that the added bodies produce only small changes to the propagated comet rotations in those configurations, without establishing that a point mass must supply the entire catalogued rotation: most of that rotation is already reproduced by the standard planetary baseline. Mass inversions that require an extra body to generate that whole angle are response-scale exercises, not measurements of a missing mass.

A point mass does not necessarily produce a detectable net orbital-energy difference whenever it deflects a trajectory. A static conservative potential can change direction while preserving asymptotic energy; stationary metric dynamics likewise admits a conserved energy associated with time-translation symmetry. Thus “rotation without detected energy contrast” is not a unique discriminator between gravity and a temporal boundary. Energy-equivalence claims require confidence bounds and an explicit alternative's predicted effect size, rather than a failure to reject a difference.

The same care applies to drag, outgassing and Galactic tides. The direct tide integrations test a specified force tensor over the simulated segments; the latitude-conditioned sky null is a separate population reference and is not a full Galactic-tide population model. Fits without nongravitational terms can absorb omitted cometary recoil into orbital elements. Null correlations with selected covariates narrow explanations without excluding all their interactions or establishing the physical absence of a force.

6.2 A temporal-boundary model must predict observables

A concrete TEP test must supply a field configuration $\phi(\mathbf{x}, t)$, the functions A and B , boundary conditions and the environmental response allowed by the canonical theory. It must then predict the matter trajectories and recorded measurements. A spatially varying conformal factor generally changes the matter equations of motion; it cannot simply be inserted as a phase offset while holding every other sector fixed by assumption.

The distinction is especially clear for an integrability claim. For a smooth, single-valued conformal factor, $\Sigma = d \ln A$ is exact and $\oint \Sigma = 0$ on a closed loop. Nonzero residual synchronization holonomy requires the additional transport structure specified in the canonical framework. The catalogue tests here do not measure a closed synchronization loop, and a static scalar gradient alone does not demonstrate non-integrable time.

The mathematical bridge to the observations can be stated without inventing a measured lapse. Let $\mathbf{y}$ contain astrometry or clock observables, $\boldsymbol{\eta}$ the conventional orbit and calibration parameters, and $\boldsymbol{\alpha}$ a finite set of physical TEP parameters. A model predicts

$$\mathbf{y} = \mathbf{f}_{\text{GR}}(\boldsymbol{\eta}) + \mathbf{s}(\boldsymbol{\eta}, \boldsymbol{\alpha}) + \boldsymbol{\epsilon}.$$

For a locally linear conventional fit with design matrix J and inverse noise covariance W , the fitted nuisance parameters absorb the component $P\mathbf{s}$, where

$$P = J(J^T W J)^{-1} J^T W,$$

with an appropriate generalized inverse if needed. The surviving signature is $(I - P)\mathbf{s}$. The same projection must be used in an injection-recovery test. This explains why a fitted orbital-element change cannot be identified with an accumulated proper-time offset without deriving the signal $\mathbf{s}$ and its absorption.

A successful model would connect the resident distribution and comet observables with shared parameters, explain the sign and temporal limits of each response, and remain compatible with clock and spacecraft constraints under their actual measurement procedures. The present analysis does not contain that completed derivation; its contribution is to identify the observational patterns and failure modes such a derivation must confront.

6.3 A falsifiable protocol

The next test should freeze the physical and statistical model before inspecting a new sample. The protocol must state the training objects, reference frame, axis, cap or continuous profile, orbital cuts, survey-selection function, observation-quality requirements, integration failures, primary outcome, effect direction and stopping rule. Changing any one of them after examining the validation set creates a new model requiring another validation sample. A temporal split in already available data is useful, but should not be represented as an externally registered future prediction without the corresponding record.

For the resident channel, apply the characterized survey simulator to both an intrinsic isotropic population and the proposed field-shaped population. Include magnitude, motion, sky coverage, tracking efficiency and the secure-orbit cut. Deduplicate objects across catalogues. Fit common parameters on training data and compare predictive likelihoods on the withheld discoveries. If the claimed excess follows survey selection after this calibration, the intrinsic-sector interpretation fails.

For the comet channel, simulate observations from both the conventional and specified TEP trajectories at the actual station epochs, with catalogued uncertainty and plausible correlated astrometric errors. Pass both through the identical orbit fitter and selection rules. Demonstrate recovery of injected signals, coverage of uncertainty intervals, false-positive calibration and convergence with integration tolerances and boundary extraction. The primary comparison must be at the observation level or a derived statistic with its validated covariance, not a reduced regression on a quantity generated by standard dynamics.

The modern-catalogue control should be part of this protocol, with composition-by-sky and station-by-epoch interactions tested or marginalized. A revised moving or bipolar boundary must predict its lobe locations and transition times from training data; merely fitting the observed reversal is insufficient. If the frozen prediction fails again, that model should be rejected without moving the target.

6.4 Forecasts are conditional

The resident cap fraction of roughly 0.59 and the adopted footprint baseline of roughly 0.42 motivate a prospective sample-size calculation (steps 054 and 067). They are measured or fitted sample fractions under the current selection assumptions, not a parameter-free TEP prediction for another survey. A future survey's expected fraction and power must be recalculated using its own selection function and uncertainty in the training estimates. No calendar date or discovery yield makes the test decisive before that calibration.

These requirements strengthen the testability of TEP: they identify a finite set of observations that could support or falsify a specific field model. They also retain the persistent spatial result as a legitimate target for conventional population models. The same quantitative standard should apply to both.

7. Conclusions

The public catalogue record contains a preferred resident sector and a persistent comet-aphelion dipole under the tested angular nulls. These are the strongest reasons to investigate a common spatial structure in the outer Solar System. Historical cometary rotation contrasts add information about the selected population, but the rotations themselves are reproduced by standard planetary dynamics and the equivalent-time conversion is not a proper-time measurement.

The temporal rotation holdout fails the original fixed-sector excess prediction. The later displaced structure is sensitive to astrometric-catalogue composition, while corrected mirror and meridian comparisons remain exploratory. Overlap among catalogues, directions from the same orbit, and shared measurement procedures precludes counting the reported channels as independent proofs.

TEP offers a possible organizing hypothesis whose value would lie in predicting these channels jointly from a specified field and matter response. The present results motivate that test but do not yet establish a temporal domain boundary or a preference for TEP over a calibrated conventional model. The decisive advance is a successful, frozen prediction of new observations through the same survey-selection and orbit-fitting procedures used to evaluate the alternatives.

8. References

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9. Data availability and reproducibility

9.1 Data record

Raw catalogue snapshots are stored under `data/raw/` ; source-specific provenance records include retrieval details and, where recorded, hashes. CODE is an archival snapshot rather than a successful live export. MPC observations and SBDB fit seeds are cached for the independent refits. Spacecraft data include NAIF kernels, PDS Voyager plasma-wave records and the Whiting & Woerner RTG dataset. Acquisition steps may update live sources; preserving the existing snapshot is necessary for reproducing the present numerical record.

The observed-data evidence is separate from Monte Carlo, bootstrap, integration and power controls. Unavailable diagnostics must not be treated as zero measurements or successful tests. The repository-wide audit inventories nonfinite fields and flags source/result provenance uncertainty rather than silently converting missing results into favourable evidence.

9.2 Running and auditing the analysis

`scripts/run_all.py` lists the analysis order and runs each step in a subprocess, stopping on failure. The claims audit runs after its upstream results. Every registered step and source file is included in the review inventory; the review report records which numerical analyses were actually rerun. A successful syntax check or a result newer than its script does not by itself demonstrate numerical or scientific validity.

`scripts/audits/audit_tep9_observables.py` independently checks frames, angular arithmetic, direct versus reduced dynamical residuals, the equivalent-time counterexample and the era dipole calculation. Focused regression tests exercise the probability contracts, cone-band union, state ordering and optimiser rejection behaviour. These tests use synthetic mathematical controls, clearly separate from empirical evidence.

The publication source is `site/components/`. Headline quantitative claims bind to exact JSON leaves through `site/claims.json` and `site/claims.js` ; invalid, missing or nonfinite bound values stop the build. Running `npm run build:markdown` in `site/` builds the HTML and Markdown from this source. The PDF is generated from the built HTML. The older wording is preserved in `reviews/pre_review_components/` for comparison and is not part of the current publication.

9.3 Reproducibility limits

Several cross-channel steps read sibling TEP projects rather than self-contained data in this repository. Their availability and source hashes must be recorded; the present paper does not claim a clean-room rerun of those entire external pipelines. Fixed random seeds do not guarantee identical floating-point results across library versions. The review records the active package versions separately from the declared requirements.

The analysis is not yet a complete generative physical model or a fully selection-calibrated significance test. Full covariance propagation, astrometric systematics, continuous encounter/boundary convergence, and independent validation of a specified TEP signal remain outstanding scientific work. The review report distinguishes repaired implementation errors from these unresolved requirements so that an existing limitation cannot be mistaken for a completed validation.